

Shakir Suleimanov

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Groningen, the Netherlands

Research interests: Single-cell atlasing & integration • Lung diseases • Cancer immunology

BIO

I am a RESPIRE-EXCEL MSCA DN PhD student in Martijn Nawijn's lab, focusing on integrating and atlasing single-cell data. My research explores disease-associated cell states, trajectories, and interactions in chronic inflammatory airway diseases. I have a background in both wet-lab and computational immunology and enjoy teaching programming and data analysis to biologists. Outside the lab, I find peace and inspiration in traveling and mountaineering.

EDUCATION

University of Groningen (PhD)

Computational research: Supervision by Martijn Nawijn and Malte Lücken

- Single-cell data integration and atlasing to study lung diseases

Groningen, the Netherlands

Oct 2025 - Oct 2028

Skoltech (MSc)

Computational research: Supervision by Dmitriy Chudakov

- Atlasing of scRNA/TCR-seq datasets of CD4+ T-cells from lung cancer

Moscow, Russia

Sep 2023 - Jun 2025

Sechenov University (MD)

Wet-lab research: Supervision by Irina Vlasova

- Characterization of human macrophages at different polarization states

Moscow, Russia

Sep 2017 - Jun 2023

ADDITIONAL EDUCATION

Bioinformatics for Biologists

Institute of bioinformatics

Moscow, Russia

Sep 2023 - Jun 2024

Biostatistics and medical data analysis

Institute of bioinformatics

Moscow, Russia

Sep 2022 - Jan 2023

SELECTED RESEARCH EXPERIENCE

Study of TIL CD4+ T-cells in KRAS-mut NSCLC

IBCh RAS, Advisor: Dmitriy Chudakov, Daniil Lukianov

- Processed raw GEX and TCR datasets, performed data processing, integration, and atlas construction.
- Conducted compositional, gene module co-expression, plasticity, and clonality analysis.

Jan 2025 - Jun 2025

Development of an automated pipeline for QC of immune repertoire sequencing

MiLaboratory, Advisor: Mikhail Myshkin

- Developed Python and R scripts for MiXCR execution, data processing, and visualization.
- Automated report generation in $\LaTeX$ and integrated all components into a single Jupyter Notebook.

Jun 2024 - Oct 2024

Investigation of Empty Droplets detection in SUMseq

Bioinformatics Institute, Advisor: Maxim Kholmatov (EMBL)

- Writing core functions for quality check, plotting, and statistical analysis for benchmarking in R.
- Running emptyDroplets on raw data. Analysis of correlation between log-probabilities of emptyDroplets with the sample coverage.

Feb 2024 - Jun 2024



Identification of genes associated with immune response using open database ImmuneSpace

Bioinformatics Institute, Advisor: Bakin Evgeny

- Exploratory analysis of vaccination dataset and data cleaning and filtration.
- Identification of DEGs using pairwise-comparisons with Mann-Whitney test.

Nov 2022 - Jan 2023



Study of functional activity of human macrophages and effect of MSC secretome on their polarization 2021 - 2023

Laboratory of Clinical Smart Nanotechnologies, Advisor: Vlasova Irina I., Kosheleva Nastasya V.

- Human PBMC isolation, differentiation and polarization.
- Analysis of functional activity with chemiluminescence, expression of markers with flow cytometry, and phagocytic activity with fluorescence and confocal microscopy.
- All statistical data analysis and fluorescence image processing

PUBLICATIONS AND PATENTS

- [2025] **Single-cell multi-omics reveals the TNF- α activation threshold for Classical Monocytes by studying healthy donors and rheumatoid arthritis patients.**
Perik-Zavodskii R., ..., *Suleimanov S.*, et al. *Frontiers in Immunology*, 16, 2025
- [2025] **Molecular Signatures of Dendritic Cell Activation upon TNF Stimulation: A Multi-Omics Study in Rheumatoid Arthritis.**
Alshevskaya A., *Suleimanov S.*, Elizaveta S., et al. *International Journal of Molecular Science*, 2025, 26(13), 6071
- [2024] **Radical-generating activity, phagocytosis and mechanical properties of human macrophages.**
Suleimanov S., Efremov Y., Klyucherev T., et al. *International Journal of Molecular Science*, 2024, 25(3), 1860
- [2023] **MSCs' conditioned media cytokine and growth factor profiles and their impact on macrophage polarization.**
Peshkova M., Korneev A*, *Suleimanov S.**, et al. (*equal contribution). *Stem cell research & therapy*, 14, 142, 2023
- [2023] **Method for single-walled carbon nanotubes cleaning.**
Irina I. Vlasova, *Shakir K. Suleimanov* et al. Rospatent №2023103987/28(008724). *Patent*
- [2022] **Redox-Activation of Neutrophils Induced by Pericardium Scaffolds.**
Irina I. Vlasova, *Shakir K. Suleimanov* et al. *International Journal of Molecular Sciences*, 2022, 23(24), 15468
- [2022] **Prevalence and risk factors of post-COVID-19 condition in adults and children at 6 and 12 months after hospital discharge: a prospective, cohort study in Moscow (StopCOVID).**
Pazukhina E., ..., *Suleimanov S.*, et al. *BMC Medicine*, 20, 244, 2022
- [2021] **Stop COVID Cohort: An Observational Study of 3480 Patients Admitted to the Sechenov University Hospital Network in Moscow City for Suspected Coronavirus Disease 2019 (COVID-19) Infection.**
Munblit D., ..., *Suleimanov S.*, et al. *Clinical Infectious Diseases*, 73, 1, 1-11, 2021

CONFERENCES

- [2024] **Multimomics single-cell analysis of gene signatures in immune cells upon activation of TNF receptors.**
Sechenov International Biomedical Summit, Presentation, Moscow, Russia
- [2024] **Automatization of data analysis in the quality control process.**
Skoltech Industrial day 2024, Poster, Moscow, Russia
- [2023] **Redox activity and morphological features of activated human M1 and M2 macrophages.**
Medical Spring, Presentation, Moscow, Russia

SKILLS

Programming: Proficient in *R*; good knowledge of *Python*; comfortable with *Bash*, *Git*, and HPC server

scRNA-seq analysis: Raw data processing (*CellRanger*); *Scanpy*, *Seurat*, *Bioconductor*, *OmicsVerse*; multimodal analysis (CITE-seq, scTCR-seq); pre-processing pipelines; vertical and horizontal integration; differential gene and pseudobulk analysis; gene set enrichment; compositional analysis; trajectory inference.

Wet lab: Cell culturing; fluorescence and confocal microscopy; flow cytometry; ELISA; chemiluminescence.

Languages: English (fluent), Russian (native).

HONORS AND AWARDS

Government scholarship for excellent students	2017, 2022, 2024
Award of Sechenov University President for scientific achievements	2022
Academic scholarship for excellent academic and scientific achievements in Sechenov University	2020-2023
Winner of personal student grant (\$5000) for scientific research	2020
Laureate of Russian Biology Olympiad	2016

TEACHING AND MENTORSHIP

Statistics & R course, Bioinformatics Institute, Teaching assistant	2024-2025
Lecture on Introduction to NGS for 1st year MSc students at Sechenov University	2023
Senior and Junior tutor at the Scientific Career's Center (mentored 4 students for 1 year)	2018-2020